

ALBANY, OH, USA

WMO: 720397

Lat: 39.211N

Lon: 82.231W

Elev: 233

StdP: 98.55

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 00131

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-12.4	-21.0	0.6	-13.3	-17.5	0.8	-10.4	10.7	1.0	9.0	3.1	1.3	270	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.7	22.0	31.5	21.7	30.2	21.4	24.6	29.0	23.9	28.3	23.1	27.3	2.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	18.8	25.8	22.6	17.9	25.3	22.0	17.1	25.0	75.7	29.0	72.5	28.4	69.6	27.4	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.1	5.7	-20.0	34.0	4.9	1.9	-23.5	35.3	-26.4	36.4	-29.2	37.5	-32.8	38.8		
(5)			-20.4	25.6	4.6	1.1	-23.7	26.4	-26.5	27.0	-29.1	27.6	-32.4	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	-1.4	0.7	6.1	12.3	17.8	21.8	23.4	22.8	19.6	12.8	6.0	2.1		
	DBStd	10.00	6.73	6.67	5.97	5.00	4.51	3.12	2.77	2.82	3.89	5.03	5.00	5.32		
	HDD10.0	1228	358	268	152	33	2	0	0	0	27	138	250			
	HDD18.3	2837	612	495	382	187	67	7	1	2	31	183	369	502		
	CDD10.0	1984	4	7	30	103	245	356	416	397	287	114	19	7		
	CDD18.3	551	0	0	1	7	51	113	158	140	68	12	0	0		
	CDH23.3	5045	0	2	10	132	487	1024	1360	1225	658	144	3	0		
	CDH26.7	1695	0	0	1	26	133	348	493	427	231	36	0	0		
(14) Wind	WSAvg	1.9	2.5	2.6	2.5	2.5	1.8	1.6	1.2	1.1	1.3	1.7	1.9	2.2		
(15) Precipitation	PrecAvg	1065	85	72	101	88	117	103	107	86	82	68	74	85		
	PrecMax	1293	153	149	201	146	208	232	147	135	256	192	142	203		
	PrecMin	848	38	22	34	32	12	22	62	32	19	16	33	43		
	PrecStd	130	32	33	42	33	60	46	23	31	57	41	27	40		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	20.8	25.9	29.8	32.0	33.0	34.9	34.8	33.0	31.2	22.8	18.9		
		MCWB	13.3	14.1	16.3	16.3	20.0	21.7	22.0	23.1	21.3	21.4	15.4	14.9		
	2%	DB	15.1	17.3	22.0	27.1	29.8	31.9	32.5	32.4	31.4	27.0	19.9	15.9		
		MCWB	12.1	12.2	14.1	16.0	20.2	21.4	22.7	22.1	21.5	18.1	13.4	13.2		
	5%	DB	11.9	14.3	19.0	24.5	28.1	30.2	31.1	30.9	29.5	24.6	17.4	13.2		
		MCWB	9.7	11.0	12.2	14.9	19.5	21.1	22.4	22.0	21.2	17.4	12.1	10.9		
	10%	DB	8.7	11.9	16.1	22.2	26.3	28.8	29.7	29.0	27.4	22.0	15.1	11.1		
		MCWB	6.3	8.4	10.7	14.1	18.9	20.6	22.0	21.4	19.2	16.3	10.9	9.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	16.1	17.2	19.1	23.1	24.3	25.8	25.4	24.1	22.4	17.3	16.0		
		MCDB	16.5	20.2	23.5	25.2	28.3	30.0	29.7	29.6	28.9	26.5	21.6	17.8		
	2%	WB	12.6	13.8	15.6	17.7	21.8	23.1	24.9	24.5	23.0	20.9	15.1	13.7		
		MCDB	14.9	15.7	19.9	22.7	26.4	28.0	29.1	28.8	27.9	24.8	17.4	15.2		
	5%	WB	9.9	11.9	14.1	16.5	20.8	22.6	24.1	23.6	22.3	19.2	13.1	11.7		
		MCDB	11.7	14.3	17.2	22.5	25.3	27.4	28.5	27.8	26.8	22.5	15.9	13.1		
	10%	WB	6.3	8.2	12.1	15.2	20.0	21.9	23.2	22.8	21.4	17.2	11.3	9.3		
		MCDB	8.5	11.1	15.1	20.0	24.6	26.7	27.3	26.7	25.1	20.8	14.8	11.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	10.3	11.7	13.7	12.0	11.7	11.0	11.7	12.7	12.6	11.8	8.8		
		MCDBR	12.9	14.3	16.3	17.5	14.2	13.2	12.3	13.4	14.7	15.7	15.5	12.4		
	5% WB	MCWBR	10.1	10.8	9.4	8.3	6.2	5.3	4.9	5.4	5.9	7.8	9.8	10.4		
		MCWBR	11.9	13.7	12.1	13.7	11.2	11.0	10.6	10.7	11.8	12.3	11.8	11.3		
(40) Clear-Sky Solar Irradiance	taub		0.310	0.324	0.352	0.384	0.427	0.448	0.465	0.464	0.395	0.365	0.328	0.302		
	taud		2.470	2.409	2.393	2.322	2.218	2.199	2.169	2.153	2.326	2.412	2.489	2.519		
	Ebn at Noon		864	898	905	892	856	835	816	807	846	838	834	849		
	Edh at Noon		79	96	109	124	141	144	147	146	115	95	78	71		
(44) All-Sky Solar Radiation	RadAvg		1.72	2.46	3.55	4.77	5.46	6.01	5.72	5.39	4.45	3.13	2.18	1.48		
	RadStd		0.17	0.35	0.31	0.42	0.46	0.38	0.32	0.32	0.43	0.32	0.28	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+120	+64

Nomenclature: See separate page